

Detection of Cyber-Attacks and Power Disturbances in Smart Digital Substations using Continuous Wavelet Transform and Convolution Neural Networks

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ABSTRACT

Smart Digital Substations use communication networks to exchange the information among its components to perform monitoring and control, which makes such components prone to cyber-security threats. The presence of power quality disturbances may add complexity to the problem of detecting such cyberattacks as some of these power quality disturbances behave in a similar manner to some attacks. In this paper, a novel approach that detects cyberattacks from power quality disturbances and normal operation is developed. The proposed approach uses only three out of the twenty-nine features to detect such cyberattacks. The proposed approach uses the continuous wavelet transform to represent such features in the time-frequency scalograms, which are then fed to a convolution neural network for detecting cyber-attacks and power disturbances in IEC-61850 substation systems. The proposed approach has been tested on three datasets from substation test systems as well as in real-time using OPAL-RT. The results have shown that the proposed approach was effective in detecting the cyber-attacks from power disturbance and normal operation with a detection accuracy of 100% in simulation environment and 99.45% in real-time using OPAL-RT.

1. Introduction

1.1. Research Motivation

The Smart Digital Substation (SDS) typically converts the binary status and the analogue measured data into digital data. Furthermore, digital data are used for information exchange among Intelligent Electronic Devices (IEDs), switchgear devices and Supervisory Control and Data Acquisition (SCADA) systems. Such information exchange is managed through communication protocols, which require the integration of communication networks. The IEC-61850, which is an interoperability standard for information exchange, plays a crucial role in enhancing the coordination of such devices within the smart digital substations and their assets. The IEC-61850 standard uses Sample Value (SV) and Generic Object-Oriented Substation Events (GOOSE) to enable seamless communication among the SDSs and smart grid assets, thus improving the system's operational efficiency. However, the use of communication networks for information exchange makes such SDSs prone to cyber-attacks. The exploitation of these vulnerabilities poses a significant threat to the secure operation of the smart grid

infrastructures. In 2016, a significant cyberattack took place in Kyiv, the Ukrainian capital. The cyberattack resulted in disabling an electric substation and leaving customers in parts of Kyiv without electricity for nearly an hour [1]. In addition to cyberattack threats, power grid disturbance events encompass a wide range of scenarios that disrupt the normal operation of a power grid such as breaker failures, busbar protection trips and under-frequency problems. For example, the blackout that occurred in the USA and Canada in 2003, were significant events that had a huge economic impact with estimated costs of nearly \$10 billion USD in the United States, and \$2.3 billion CAD in Ontario, Canada [2].

1.2. Innovation Aspects

The work presented in this study aims to address the problem of cyberattacks detection in smart digital substations by developing a novel approach that uses the least number of features (i.e., three out of twenty-nine in this study) to perform the detection of the attacks from power disturbances and normal cases. It is worth noting that the similarity between the behaviour of some cyberattacks and power disturbance events makes the cyberattack detection problem a challenging task.

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Nomenclature*Abbreviations*

SDS	Smart Digital Substation.
IEDs	intelligent electronic devices.
SCADA	Supervisory Control and Data Acquisition.
SV	Sampling Value.
GOOSE	Generic Object-Oriented Substation Events.
CWT	Continuous Wavelet Transform.
CNN	Convolutional Neural Network.
FDIA	False Data Injection Attacks.
ML	Machine Learning.
LSTM	Long Short-Term Memory.
NIDS	Network Intrusion Detection System.
MVCC	Minimum Voting for Critical Class.
ARLS	Alternate Recursive Least-Square.
GWO	Grey Wolf Optimization.
PSVM	Proximal Support Vector Machine.
GCN	Graph Convolutional Network.
DoS	Denial of Service.
LSTM	Long Short-Term Memory.
PSO	Particle Swarm Optimization.
BiLSTM	Bidirectional Long Short-Term Memory.
ReLU	Rectified Linear Unit.
BusF	Busbar Failure.
BF	Breaker Failure.
UF	Under-Frequency.
MS	Message Suppression.
DM	Data Manipulation.
ComA	Composite Attacks.
FL	Fixed Loading.
VL	Variable Loading.
Stnum	Status Number.

Sqnum	Sequence Number.
FPGA	Field Programmable Gate Array.
GE-UR	General Electric Universal Relay.
GNSS	Global Navigation Satellite System.
GUI	Graphic User Interface.

Parameters

b	Wavelet time shift.
n	Normalizing constant
ω	Angular frequency
τ	Decay and the compactness parameter
γ	Wavelet symmetry
I	Convolution input patch
P	Pooling window
α	Flatten layer Vector
Λ	The number of classes

Variables

$C(a,b)$	The coefficients of wavelet transform.
$f(t)$	The original signal.
$H(\omega)$	Heaviside step function
p^2	Wavelet Time-bandwidth
$\Psi(\omega)$	Mother wavelet.
w	Convolution filter weights
N	Convolution operation output
$R(N)$	ReLU output
G	Max pooling output
M	Flatten layer weight matrix
S	Flatten layer score
AC	Classification accuracy
CP	Number of observations correctly predicted.
RP	Total number of observations

Therefore, there is a dire need to develop a detection approach that is capable of automatically learning the key relevant features of the cyberattacks, power disturbances and normal operation without being prone to overfitting. Furthermore, it is vital to adopt the most suitable observation window during the feature extraction process to ensure the best performance and hence achieving the highest detection accuracy to eliminate the false positive/negative rates.

1.3. Contributions

The contributions of this paper are described as follows:

- The development of an effective method to detect cyberattacks from power quality disturbances and normal operation applied to smart digital substations.
- The identification of three key features out of the twenty-nine features that strongly influence the cyberattack detection in smart digital substations and use them to detect the cyberattacks and the power disturbances.
- Investigating the effect of different types of observation windows on the cyberattack detection problem as well as identifying the most suitable window that improves the detection accuracy.
- The proposed approach uses the Continuous Wavelet Transform (CWT) to generate scalograms in the time-frequency domain, which are then fed to the Convolutional Neural Network (CNN) to automatically learn the salient features from the scalograms and use them in the classification process.
- The proposed approach has gone through a set of rigorous testing and the results have shown that it can achieve 100% detection

accuracy of the cyberattacks from power disturbances and normal operation.

- The proposed approach has been tested in real-time using OPAL-RT and the results have shown that it can successfully detect the cyberattacks with a low computational time.

1.4. Addressing the Sections of the Manuscript

This paper is organized as follows: Section 2 describes the previous work developed in the literature to address the problem of detecting cyber-attacks and power disturbances. Section 3 presents the use of an enhanced signal processing technique, namely CWT along with a custom CNN to provide an effective detection approach. Section 4 presents the implementation results of the proposed approach on publicly available datasets and a real-time testing using OPAL-RT. Section 5 provides further interpretations of the detection results, comparison with other results from the literature, and discusses the computational complexity. Finally, the conclusion and future works are presented in section 6.

2. Literature Review

2.1. Previous Work

In the literature, different approaches were used to detect anomalies in power systems. Studies [3] – [8] applied statistical-based approaches, while studies [9] – [14] used Machine Learning (ML) approaches, and studies [17] – [21] focused on the Deep Learning (DL) approaches.

Table 1

Comprehensive comparison with other studies in the literature.

Study	Detecting Power Disturbances	Types of Attacks Considered in the Testing	Key Features Identification	Real-time Experimental Testing
[17]	×	F DIA	×	×
[18]	×	D oS	×	×
[19]	✓	FDIA	×	×
[20]	×	F DIA, DoS, Replay	×	×
[11]	×	D M	×	×
[5]	×	D oS	×	×
[6]	×	F DIA	×	×
[7]	×	F DIA	×	×
[12]	×	D oS, Replay, MS, DM	×	×
[8]	✓	DoS, FDIA	×	×
[21]	×	F DIA, Replay	×	×
[13]	✓	DoS	×	×
[14]	✓	FDIA	×	×
Proposed Approach	✓	Dos, MS, DM, FDIA, Replay, Composite attacks	✓	✓

2.1.1. Statistical-based approaches

Elrawy et al. in [3] designed a cyber-attack detection model by comparing the behaviour of the normal scenarios with the attack behaviour using the rule-based Network Intrusion Detection System (NIDS). Jay et al. in [4] proposed an unsupervised learning-based intrusion detection system to detect attacks targeting GOOSE packets within the substation. Elbez et al. in [5] designed a detection approach for GOOSE Network Traffic (EDA4GNET). Goyel et al. in [6] applied a technique called Minimum Voting for Critical Class (MVCC) to detect False Data Injection Attacks (FDIA). Khazaei et al. in [7] used Alternate Recursive Least-Square (ARLS) estimation to detect FDIAs targeting transmission lines. Panthi et al. in [8] applied the binary Grey Wolf Optimization (GWO)-based feature selection to classify attacks in publicly available datasets. The statistical-based approaches lack the ability to automatically learn the salient features from the data without explicit programming, which increases the computational time and complexity.

2.1.2. Machine learning-based approaches

Malik et al. in [9] applied Proximal Support Vector Machine (PSVM) algorithms on IEC-61850-based substations to detect cyber-attacks, while Saini et al. in [10] investigated seven ML algorithms to detect cyber-attacks. Bhusal et al. in [11] proposed ML approach to detect falsification attacks, while Alani al. in [12] applied a ML technique to detect and classify cyber-attacks on the DNP3 intrusion detection dataset. The research work in [13] and [14] applied ML techniques to detect power disturbances and cyber-attacks on publicly available datasets. The ML-based approaches require human intervention to manually extract the relevant features from the data to use them in the classification process [15]. Furthermore, these features may vary from one attack to another, which makes such approaches very challenging to implement. Moreover, ML approaches exhibit a limitation in discerning the features and patterns associated with different attack types, especially when there is a high degree of similarity between the attacks and the normal operation. Consequently, this leads to a large number of misclassifications [16].

2.1.3. Deep learning-based approaches

Vincent et al. in [17] proposed a Graph Convolutional Network (GCN) to detect FDIAs, while Silva et al. in [18] adopted an artificial neural network to detect Distributed Denial of Service (DDoS) attacks target networks using the IEC-61850 protocol. Bitirgen et al. in [19] designed a FDIA detection approach using a combination of three techniques: CNN, Long Short-Term Memory (LSTM), and Particle Swarm Optimization (PSO). Nonetheless, the cyber-attacks in reality encompass various strategies to target power systems, but the previous studies detected only one type of attack applied to the power system. This raises a call for a proposed detection approach that is capable to discriminate multiple attacks features and patterns. Zhang et al. in [20] applied the deep capsule CNN to detect multiple cyber-attacks, while Wang et al. in

[21] used the Bidirectional Long Short-Term Memory (BiLSTM) approach to detect attacks in substation systems. These studies did not consider detecting disturbances scenarios, which behave very similar to the attacks and hence leading to increased false positive/negative rates.

2.2. Research Gaps

Table 1 provides a comprehensive comparison between the proposed approach and the most recent studies in the literature. Visual inspection of Table 1 reveals that most of the studies in the literature that looked into the detection of cyberattacks did not consider identifying the most important key features that are relevant to the cyberattacks. This makes most of the approaches developed in the literature prone to overfitting the datasets they are trained on and hence exhibit poor performance when tested on different datasets. Thus, there is a dire need for a detection approach that is capable to automatically extracting the key relevant features of the cyberattacks from the data and using them in the detection process rather than exploring such features manually. Furthermore, few studies only looked into considering the power disturbances and only few types of attacks when detecting cyberattacks in smart digital substations, which may lead to increased false positive/negative rates of these methods. It is worth noting that some disturbances cases such as breaker failures, busbar protection trips and under-frequency problems behave very similar to some attack cases and considering them in the classification process could be very challenging and might affects the detection accuracy. Therefore, it is very crucial to develop a detection approach that is able to distinguish between disturbances events and cyber-attacks. Moreover, to the best knowledge of the authors, none of the studies in the literature have tested their approaches in real-time and hence there are uncertainties regarding the performance of such approaches in real-time testing as well as the associated computational time.

3. Problem Statement and Methodology

3.1. Problem Statement

The presence of power quality disturbance events makes the process of detecting and classifying such cyberattacks from the power quality disturbances a challenging task, which leads to several cyberattacks going undetected. Since some of these power disturbances behave in a similar manner to some cyberattacks, there is a pressing need to develop a method that automatically detect and classify cyberattacks from the power quality disturbances in SDSS. Furthermore, in SDSSs, there are several physical and network features and hence it is very important to identify the key relevant features to the cyberattacks to be able to successfully detect such attacks at a very high accuracy. This issue the call for a detection approach that is capable to automatically learn and extract the key relevant features of the cyberattacks from the data by

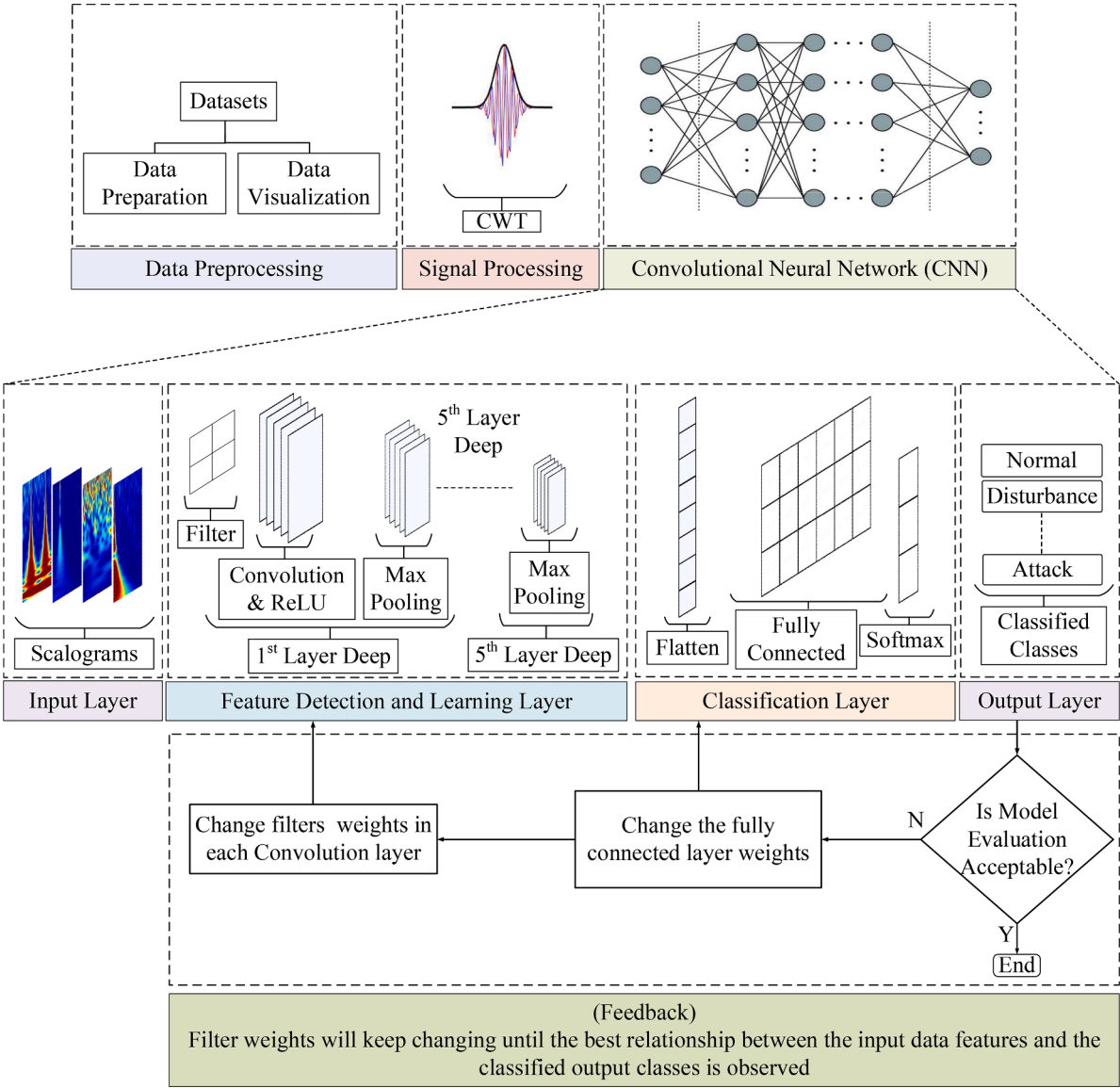


Fig. 1. The workflow of the CWT-CNN detection model.

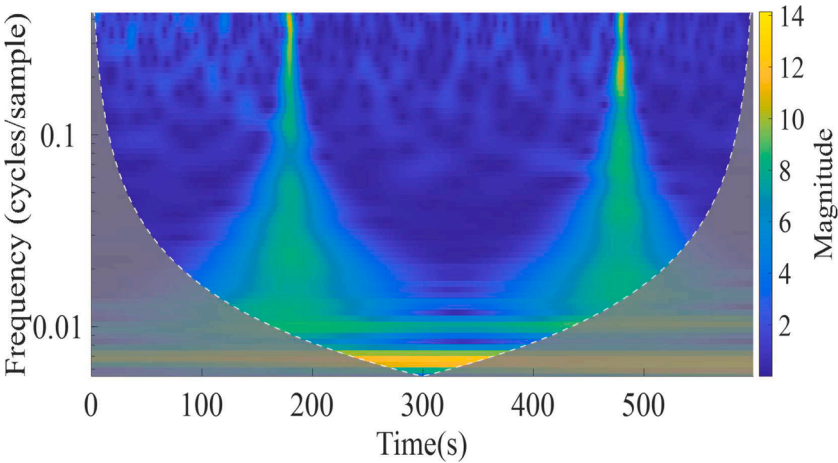


Fig. 2. Time-frequency scalogram of IED current values.

Table 2
Confusion Matrix.

		Predicted Class		
		Normal	Disturbance	Attack
Actual Class	Normal	CP	RP	RP
	Disturbance	RP	CP	RP
	Attack	RP	RP	CP

applying an advanced signal processing tool and a deep learning technique, eliminating the need for manual exploration of these features.

3.2. Methodology

The proposed approach in this work uses the CWT as the signal processing tool and the CNN as the deep learning technique. Fig. 1 illustrates the workflow of the proposed detection approach. First, the input data undergo the preprocessing stage. Typically, there are two sets of features in SDSs; physical features such as voltage, current, frequency and network features such as GOOSE data, which may include heart-beat, GOOSE length, status and sequence numbers. The CWT is applied to these features to produce the wavelet coefficients that are represented in the time-frequency scalograms (or images). These scalograms are then fed to a custom neural network called CNN to extract the features directly from these representations and use them in the classification process. Algorithm 1. summarizes the CWT-CNN detection approach.

Algorithm 1. The pseudo code of CWT-CNN
1. **Initialization**
2. **Determine the system topology and preprocess the datasets**
Data Visualization
Removing Anomalies
Estimating missing values
Replacing duplicated time stamp values
3. **Apply the CWT on the data to generate scalogram images as in Eq. (1)**
(continued on next column)

(continued)

4. **Split the dataset into training data and testing data**
5. **Training Process**
for each CNN convolutional layer = 1 to 5
Apply the convolution operation as in Eq. (3)
Apply the ReLU Activation Function as in Eq. (4)
Apply the max pooling operation as in Eq. (5)
end for
Apply the fully connected operation as in Eq. (6)
Apply the softmax operation as in Eq. (7)
If the model evaluation is acceptable
Perform step 7
Else
Change filters weights in Eq. (3) and (6) using the backpropagation operation
6. **Fix the filters weights in Eq. (3) and (6) as the CNN network is already trained**
7. **Testing Process**
for each CNN convolutional layer = 1 to 5
Apply the convolution operation as in Eq. (3)
Apply the ReLU Activation Function as in Eq. (4)
Apply the max pooling operation as in Eq. (5)
end for
Apply the fully connected operation as in Eq. (6)
Apply the softmax operation as in Eq. (7)
8. **Perform the classification**
9. **End**

3.2.1. Data Preprocessing

In this stage, a preliminary investigation of the data is performed to understand the specific data characteristics and to identify the data features, which helps in selecting the appropriate analysis and pre-processing techniques. Secondly, these signals are pre-processed to remove any anomalies, estimate missing values, and replace duplicated time stamp values. Finally, the data are visualized and inspected to detect the specific features from the data and to evaluate the most important signals to use for the classification.

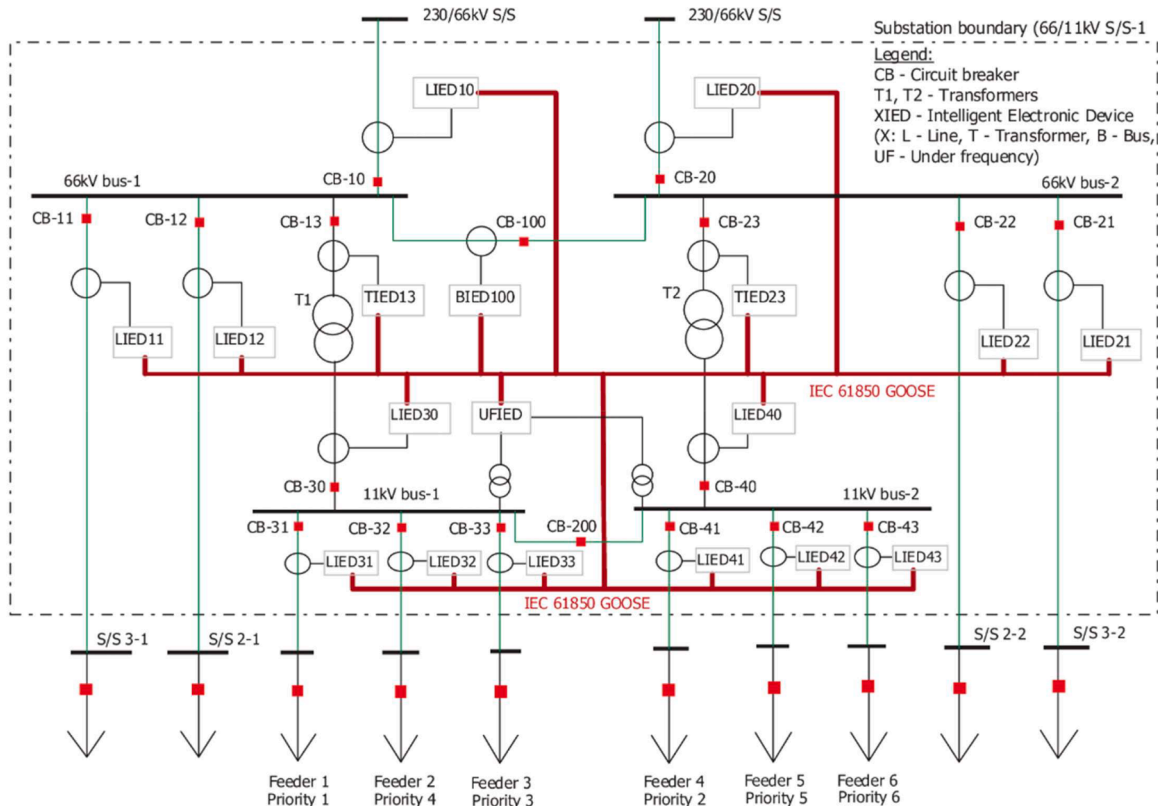


Fig. 3. IEC 16850-Based substation one-line diagram.

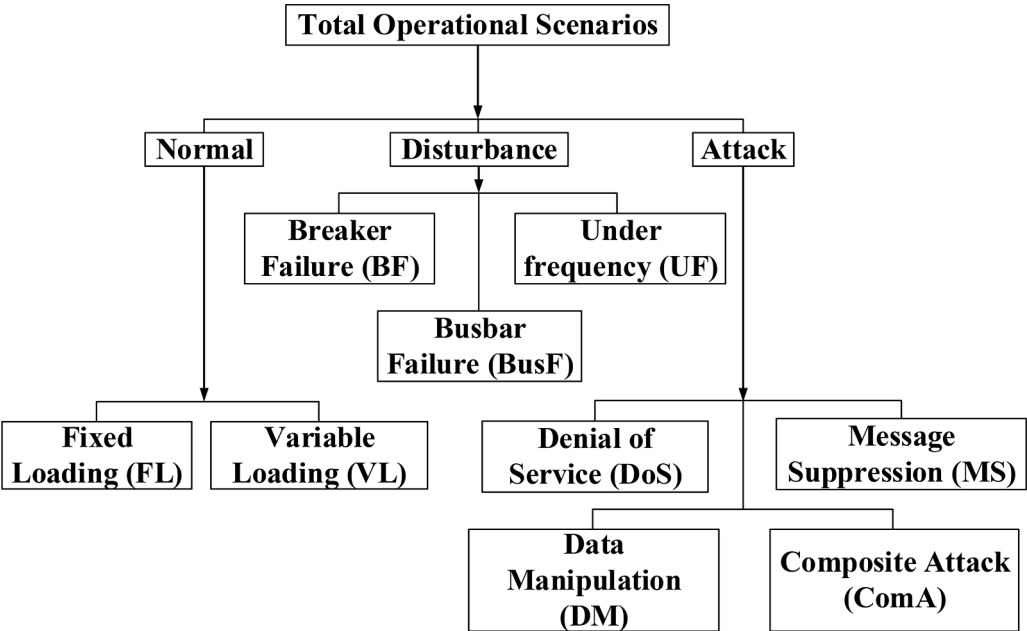


Fig. 4. Operational scenarios applied on the IEC-16850 substation system.

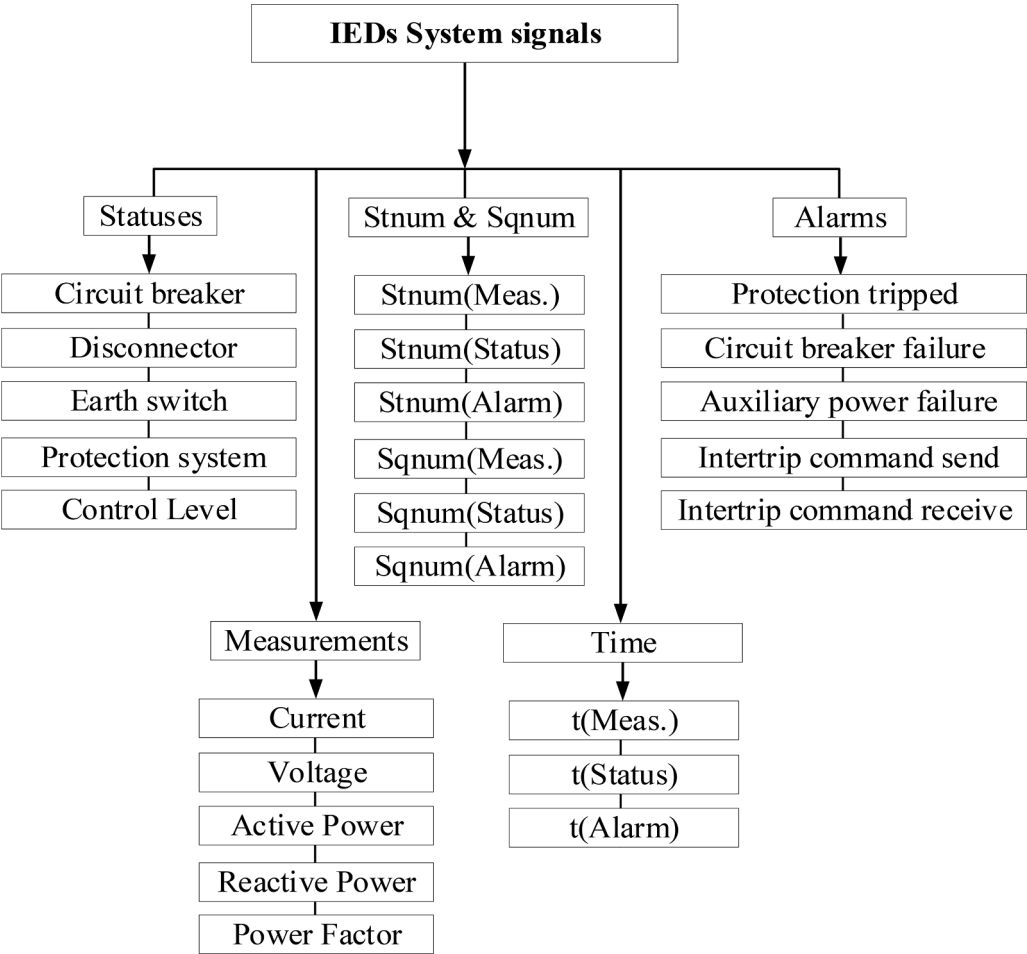


Fig. 5. The IEC-16850 substation system signals.

Table 3
Cyber-attacks Features Ranking Versus Normal and Disturbances.

			Operational Scenarios								
			Normal		Disturbance			Attack			
			FL	VL	BF	BusF	UF	DM	DoS	MS	ComA
IEDs Signals	Status	Circuit breaker									
		Disconnecter									
		Earth switch									
		Protection system									
		Control level									
	Alarms	Protection tripped									
		Circuit breaker failure									
		Auxiliary power failure									
		Intertrip command send									
		Intertrip command receive									
	Measurements	Current									
		Voltage									
		Active Power									
		Reactive power									
		Power Factor									
	Stnum & Sqnum	Stnum(measurments)									
		Stnum(Status)									
		Stnum(Alarm)									
		Sqnum(measurments)									
		Sqnum(Status)									
		Sqnum(Alarm)									
	Time	t(measurments)									
		t(Status)									
t(Alarm)											
		Strong Signal									
		Intermediate Signal									
		Weak Signal									

3.2.2. Signal Processing using the Continuous Wavelet Transform (CWT)

Signal processing is a set of techniques used to analyze, synthesize, and modify signals. Signal processing is important to observe and detect the important components of the signal [22]. The CWT is used in this work as the signal processing tool. The CWT is a mathematical technique that builds a time-frequency representation of a signal and provides a good time and frequency localization using wavelets [23,24]. The wavelet is a function that exists for a finite duration, rapidly decaying with zero means. The Morse wavelet is used in this work due to its flexibility in providing different shapes and sizes of its mother version. This makes such a function capable of generating an accurate time resolution at high frequencies and accurate frequency resolution at low frequencies [25,26]. The CWT of a signal $f(t)$ is represented as:

$$C(a, b; f(t), \Psi(t)) = \int_{-\infty}^{\infty} f(t) \frac{1}{a} \Psi^* \left(\frac{t-b}{a} \right) dt \quad (1)$$

Where, $f(t)$ is the original signal, a is the scaling factor and b is the shifting factor, $\Psi(t)$ is the mother wavelet, which represents the mother wavelet before any scaling or shifting is applied. The shifted and time-scaled wavelets are called baby or daughter wavelets [25],[27]. Finally, the CWT coefficient $C(a, b)$ is obtained by comparing the signal with the shifted and scaled versions of the daughter wavelets. The Morse

wavelet function Ψ is represented as [28]:

$$\Psi_{p,\gamma}(\omega) = H(\omega) n_{p,\gamma} \omega^{\frac{p^2}{2}} e^{-\omega^\gamma} \quad (2)$$

Where, ω is the angular frequency, $n_{p,\gamma}$ represents the normalizing constant and $H(\omega)$ is the Heaviside step function. The p^2 is the time-bandwidth, which is responsible for determining the wavelet duration in time and it is given as $p^2 = \tau\gamma$, where τ characterizes the decay and the compactness parameter, and γ controls the symmetry of the wavelet in time through the demodulate skewness [29]. In this work, the best CWT transformation results are obtained using ($\gamma = 3$) and ($p^2 = 60$), where the wavelet skewness is equal to 0, and the Morse wavelet has the minimum Heisenberg area.

Fig. 2. depicts the time-frequency scalogram of a current signal after applying the CWT. In Fig. 2, the frequency is represented in cycles per sample, while the value that quantifies the match between the wavelet function $\Psi(t)$ and the original signal $f(t)$ is represented as the scalogram magnitude and is represented by the colour code shown on the right-hand side of the figure. This makes scalograms very helpful in visualizing abrupt changes in this signal as applying CWT helps in representing changes in the time and frequency domain. The generated scalograms will be used to feed the deep learning network and train the hidden neural layers. Finally, these scalograms will be classified into output classes (i.e., normal, disturbance and attack) classes. Therefore,

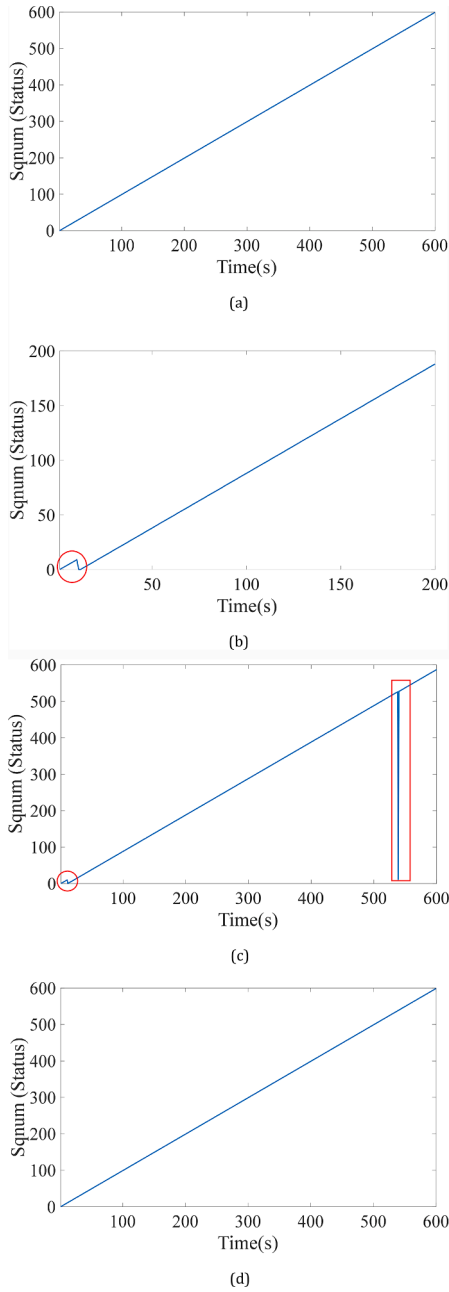


Fig. 6. Investigating the Snum (Status) feature behaviour in (a) normal scenario, (b) busbar failure, (c) denial of service attack, (d) under frequency.

the use of scalograms with clear characteristics, makes it easier for the hidden network to learn and extract the distinctive information in the signal thus improving the detection accuracy.

3.2.3. Deep Learning using Convolution Neural Network (CNN)

The deep learning is a class of machine learning techniques inspired by the human brain. What gives deep learning advantages over other machine learning techniques is its ability to automatically detect the features from the data and use them to predict future events. Thus, there is no need for human intervention to manually extract the features from the data. This expedites the detection process and decreases the computational complexity especially when dealing with large datasets. The CNN is a class of deep neural network that can be applied on any data that have a two-dimensional structure, such as images. The CNN in this paper uses a pre-trained network called AlexNet, which is capable to learn the optimal features from the new data and classify these features

during the training process into normal, disturbance and attack classes. The CNN consists of three main layers: Input layer, feature detection and learning layer and classification layer as illustrated in Fig. 1.

Input Layer

The scalograms generated by applying the CWT on the input data will be used as an input layer to feed the CNN hidden layers where salient features will be observed and detected.

Feature Detection and Learning Layer

The feature detection and learning layer contains 5 layers deep. Each layer contains three sub-layers: convolution layer, Rectified Linear Unit (ReLU) layer, and max pooling layers. In the convolutional layer, the input layer is split into patches. Each patch will pass through a filter with learning weights in an operation called convolution. The convolution is presented in the following equation:

$$N_{ij} = \sum_{r=1}^Y \sum_{s=0}^Z w_{rs} I_{(r+i)(s+j)} \quad (3)$$

Where, w is the filter weights matrix. The r and s represent the indices of w . The $(Y \times Z)$ represents w size [30], while I represents the input patch matrix. The weighted filter will be slid across the input layer patches. In every slide shift, a dot product of w and I will be performed and saved in the neuron N in the subsequent layer [31]. The i and j represent the neuron N coordinates. During the training process, the filter weights will be changed using the backpropagation operation to observe the best relationship between the input data features (i.e., input scalograms) and the classified output classes (i.e., normal, attack, disturbance). The ReLU operation is presented in Eq. (4). The ReLU output $R(N)$ will match the neuron value N in the convolution layer if it is a positive value, but if the neuron value N is negative, then the output $R(N)$ will be set at zero. The ReLU operation target is to prevent the exponential growth of the computational process during the neural network operation [32].

$$R(N) = \max\{0, N\} \quad (4)$$

The max pooling layer in (5) is used to reduce the dimensionality while preserving the most important features of the ReLU layer. This is done by applying the pooling window P . The pooling window will pass only the biggest number G in the window P . The c and l represent the pooling window dimension [33].

$$G = \max\{P_{[1][1]}, P_{[1][2]}, \dots, P_{cl}\} \quad (5)$$

These operations (i.e., convolution, ReLU and max pooling) will be repeated on the remaining 4 layers deep. The last hidden layer will be then used to feed the classification layer [15].

Classification Layer

The classification layer contains three layers: flatten, fully connected layers and softmax layers. The flatten layer is used to flatten the 2-D layer received from the feature detection and learning layer into a 1-D linear vector. In the fully connected layer, (i.e., represented in Eq. (6)), the vector α produced from the flatten layer is multiplied by a weight matrix M and added to a bias B to calculate the score S_u for each used feature, which identifies the strength of the feature among the other detected features [34,35].

$$S_u = \sum_{\eta=1}^V M_{\xi u} \alpha_{\xi} + B_u \quad (6)$$

Considering matrix M size is $(V \times L)$, then the vector α size is $(L \times 1)$, and α_{ξ} represents feature ξ in vector α . The output vector S will be $(V \times 1)$, where V represents the number of the classified output classes (i.e., normal, attack, disturbance, etc.). The matrix weights will be changing using the backpropagation operation based on the feedback from the output classes that were correctly and incorrectly predicted by the model during the training process. In the softmax layer, the softmax function in (7) converts the vector of probabilities produced from the

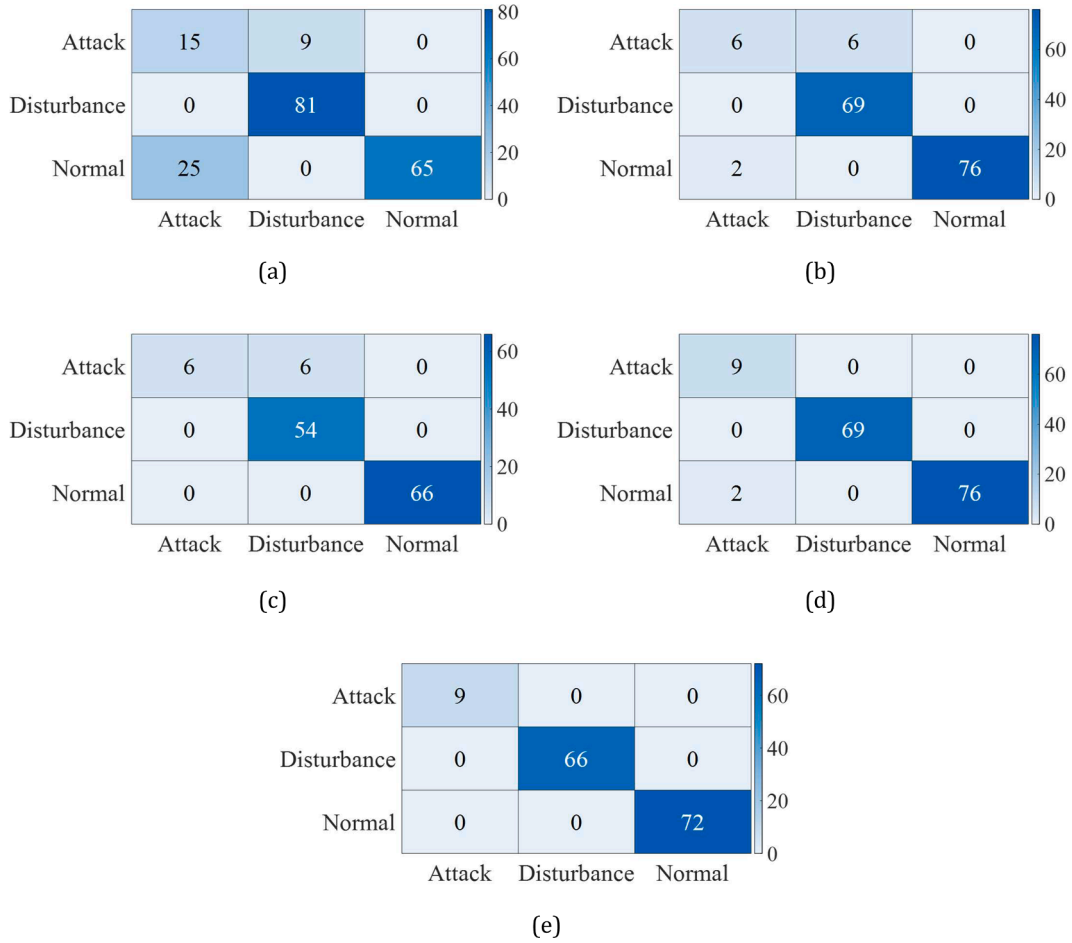


Fig. 7. Confusion matrices of using different IEDs for training and testing. (a) Simulation-1 (b) Simulation-2 (c) Simulation-3 (d) Simulation-4 (e) Simulation-5.

Table 4

Detection Results.

Simulation No	IEDs used for model training	Number of Tested IEDs	Detection Accuracy %
1	IED (10)	17	82.56
2	IEDs (10,11,22)	15	94.97
3	IEDs (10,11,22,31,32)	13	95.45
4	IEDs (10,12,22)	15	98.72
5	IEDs (10,12,22,21)	14	100

Table 5

Percentage detection accuracies using different observation windows.

None	Bartlett	Gaussian	Hamming	Kaiser
98.72	90.38	80.77	96.15	100

fully connected layer S into binary classification scores to predict the output classes.

$$Sofimax(S_\eta) = \frac{e^{S_\eta}}{\sum_{k=1}^{\Lambda} e^{S_k}} \quad (7)$$

Where, η represents the feature position in S , the Λ represents the number of classes [36].

3.3. Model Training

AlexNet has undergone training on a wide range of images, where

rich feature representations have been learned. Additionally, AlexNet is competent to classify images across 1,000 object categories [37]. This makes AlexNet capable of extracting the relevant features related to cyber-attacks and power disturbances. During the training process, the number of epochs (i.e., training process steps) was adjusted to 12, while the number of iterations was 96. The training learning rate was fixed at 0.0001 to decline the training loss on each mini-batch. The previous parameters have been identified after attempts to achieve the best training accuracy which leads to the highest test accuracy and fastest detection time.

3.4. Model Evaluation

The model evaluation reflects the strength of the network prediction for normal, disturbance and attack classes. The model evaluation is performed using the confusion matrix as shown in Table 2. The accuracy measure is used to assess the performance of CNN classifiers used for cyber-attack detection.

$$AC = \frac{\sum CP}{\sum RP} \quad (8)$$

Where, AC is the classification accuracy, CP is the number of observations correctly predicted by the model and RP represents the total number of incorrectly predicted observations.

4. Results and Evaluation

In this paper, two test systems are used in this work to evaluate the performance the proposed approach. The first test system is an 18-IEDs

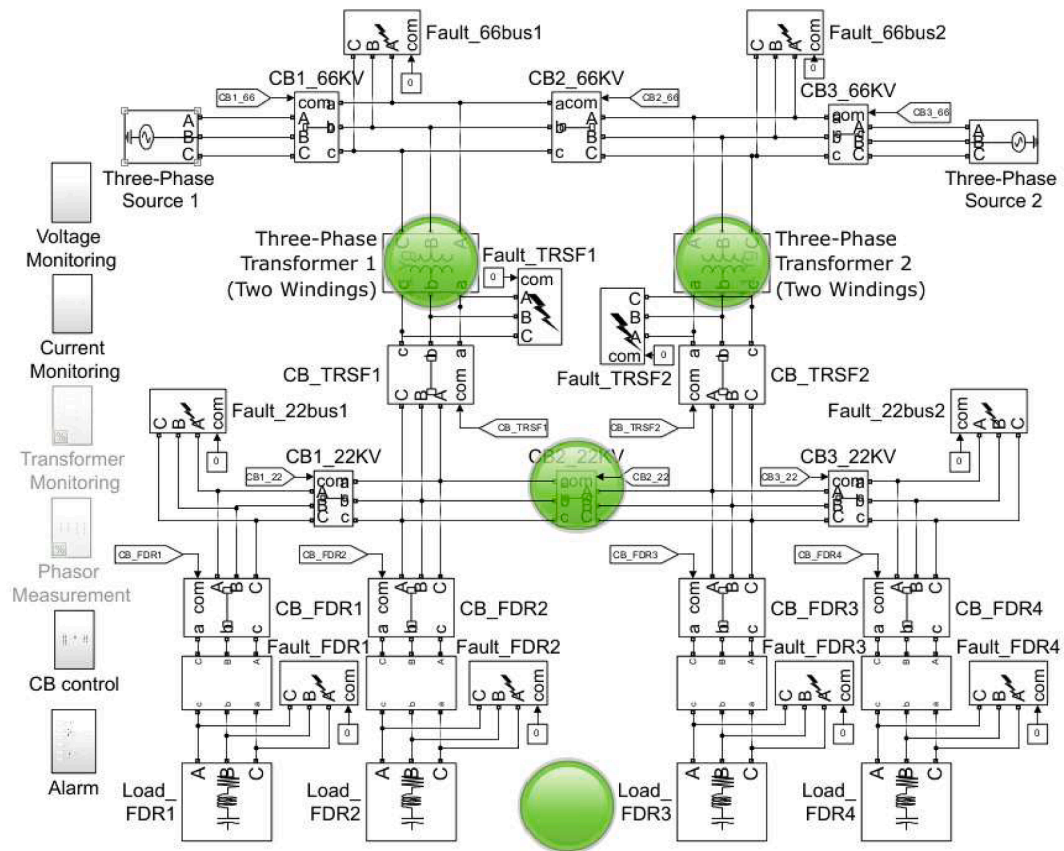


Fig. 8. The 4-IEDs substation test system simulated in MATLAB.

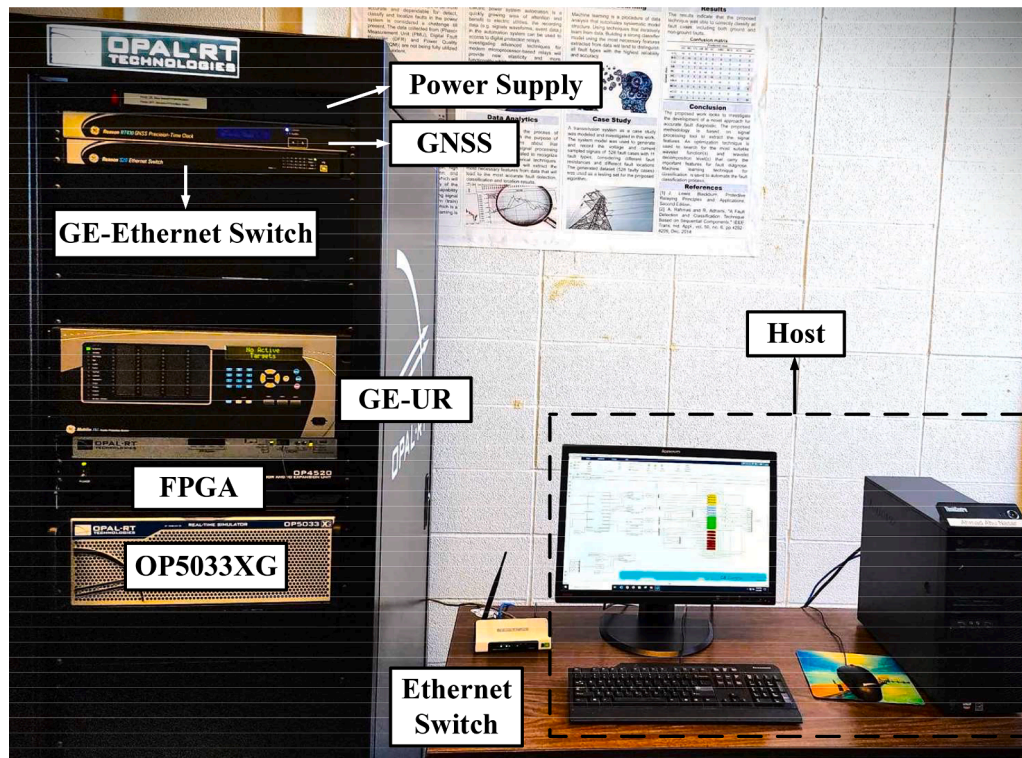


Fig. 9. The experimental set-up of the real-world test system.

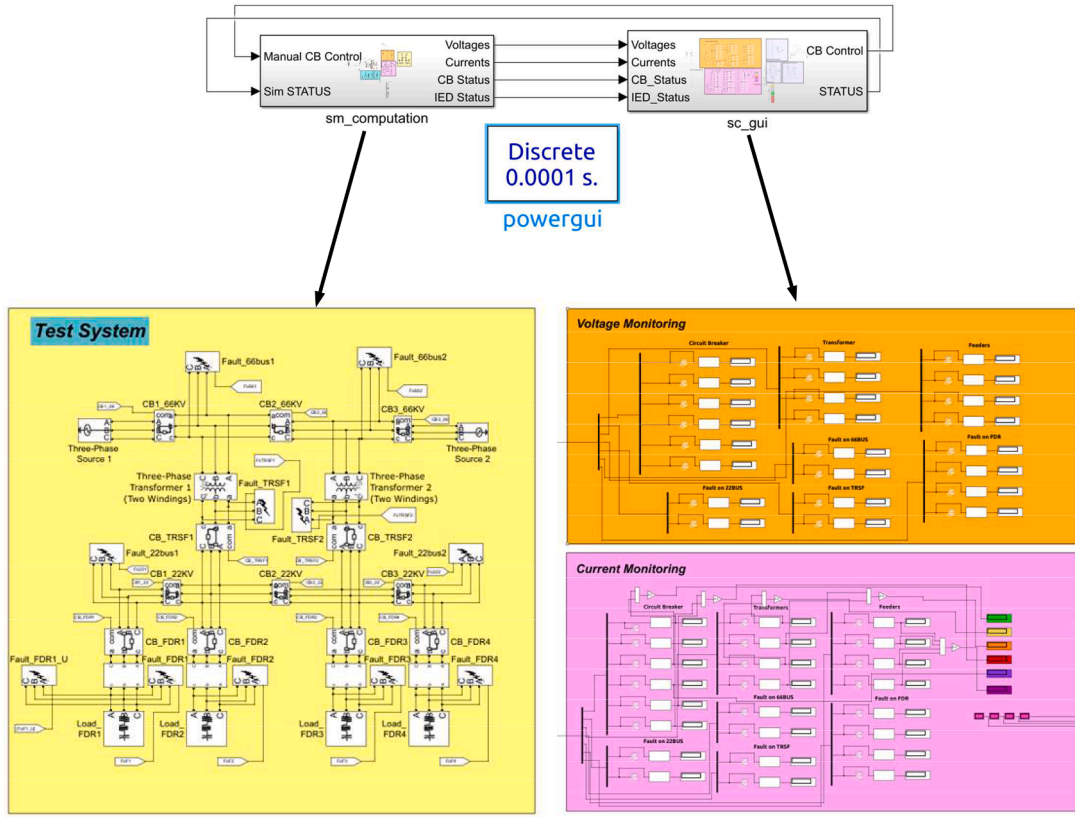


Fig. 10. System modelling in RT-LAB.

substation system [38] while the second test system is a 4-IEDs substation system [39]. In order to assess the performance of the propose approach experimentally, the 4-IED substation test system is modelled in OPAL-RT and the performance has been evaluated in real-time.

4.1. The 18-IEDs Substation Test System

The 18-IED dataset investigates cyberattacks and power quality disturbances applied to a substation automation system [38]. The test system represents a 66/11kV substation with 18 IEDs as illustrated in Fig. 3. All IEDs are communicating using the IEC-61850 GOOSE protocol. In IEC-61850 GOOSE communication, data (i.e., status, alarms, measurements, etc.) are grouped into a data set and transmitted over the electrical substation network quickly and reliably [40].

4.1.1. Operational Scenarios

In this work, three operational scenarios were applied to the test system including normal, disturbance and cyber-attack scenarios as illustrated in Fig. 4. Normal operation scenarios include fixed and variable loading situations. Disturbance scenarios consist of Busbar Failure (BusF), Breaker Failure (BF) and Under-Frequency (UF) events. Cyber-attack cases are represented by four attack types: Denial of Service (DoS) attack, Message Suppression (MS) attack, Data Manipulation (DM) attack and Composite Attacks (ComA). The DoS attack denies the information flow to the intended IEDs, while the MS attack modifies the GOOSE message and prevents IEDs from receiving specific messages. The DM attack refers to a malicious act when attackers spoof false information and subtle modifications to data for their own gain [41]. The composite attack (ComA) involves a combination of two attack types by combining different attack strategies [42].

4.1.2. Data Exploration

In this stage, the IEC-61850 substation data are collected from the

publicly available data website [38], analyzed and labelled with the appropriate classes' labels. The available data provide five categories for each IED in the IEC-16850 substation system such as statuses, alarms, time, measurements, the Status Number (Stnum) and the Sequence Number (Sqnum). The Sqnum is a counter that increments each time a GOOSE frame is sent, while the Stnum is a counter that increments every time the data changes. Under these data categories, there are 29 different signals for each IED in the IEC-61850 system as illustrated in Fig. 5. Then, these signals were pre-processed, visualised and investigated to identify to most important signals in the detection process.

4.1.3. Cyberattacks Feature Ranking for Improved Detection

All IEDs features were investigated to identify the most important features for detecting cyber-attacks and power disturbances as illustrated in Table 3. The colour code refers to the prominent features (labelled strong) in green, the non-prominent features (labelled weak) in red while the non-prominent features (labelled intermediate) in yellow. Fig. 6 is an example of investigating the feature's strength in detecting cyber-attacks and power disturbances during different operational scenarios. Fig. 6(a) depicts the Sqnum (Status) network feature under normal operation, while the small change in the Sqnum (Status) feature (i.e., pointed in the highlighted circle) indicated the Busbar Failure (BusF) event in Fig. 6(b). This small variation in the Sqnum (Status) is not very clear. Thus, the Sqnum (Status) is considered intermediate signal strength in detecting such an event and is highlighted in yellow in Table 3.

The Sqnum (Status) experienced rapid changes in the Sqnum values during the DoS attack as highlighted in Fig. 6(c). Thus, the Sqnum (Status) is considered a strong feature in detecting such attacks and is highlighted in green in Table 3. The absence of any difference between the normal scenario (i.e., Fig. 6(a)) and the under-frequency (i.e., Fig. 6(d)) made the underfrequency a weak feature to be used for detecting such events, and hence it is highlighted in red in Table 3. After

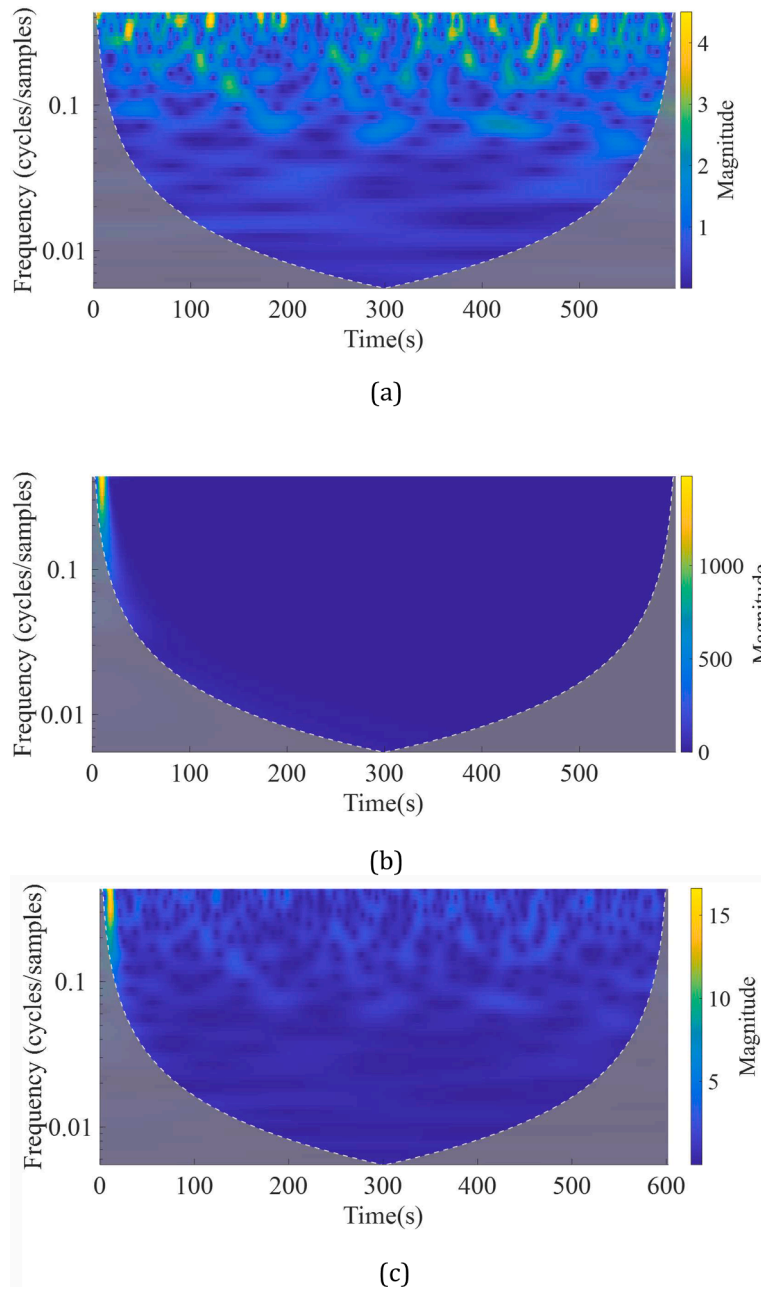


Fig. 11. Scalograms of the current signal under different operational scenarios (a) Normal-full load, (b) Disturbance-breaker failure, (c) Attack-data manipulation.

investigating the importance of all features using the detection accuracy, a combination of three features only (i.e., Current, Stnum (Status), and Sqnum (Status)) were used for detecting cyber-attack and power disturbances. This combination was found to detect and classify cyber-attacks from disturbances at 100 % accuracy as it will be demonstrated in the following subsections. This was achieved using only three features from the twenty-nine features, which resulted in a significant reduction in data pre-processing and computation complexity.

4.1.4. Detection Results

The proposed cyber-attacks and power disturbances detection approach has undergone rigorous training and testing scenarios. During the model evaluation, some IEDs data will be used to train the DL network, where the features and the patterns will be detected and observed, while the remaining IEDs data will be used to test the model's accuracy (i.e., IEDs data that has not been seen by the model before). Fig. 7 shows the confusion matrices results when using different IEDs for

training and testing. It can be observed from Fig. 7 that when IEDs (10,12,22,21) were used to train the CNN neural networks, no case was misclassified and hence the obtained accuracy is 100%. Table 4 lists the detection results of using different IEDs for training and testing. The detection results show that the use of IEDs (10,12,22,21) for training results in a detection accuracy of 100%. This is because such a group of IEDs contained the most important features from the IEDs data, which when used to train the DL network, those features were extracted and used to train the model.

4.1.5. Enhancing the Detection Accuracy by Applying Observation Windows

The time-frequency transform using CWT could produce bright spots in scalograms, which may mask some important information that exists in the time domain. In this work, the observation windows in [43] were investigated to study their effect on the detection accuracy. Table 5 lists the obtained detection accuracies when using different observation

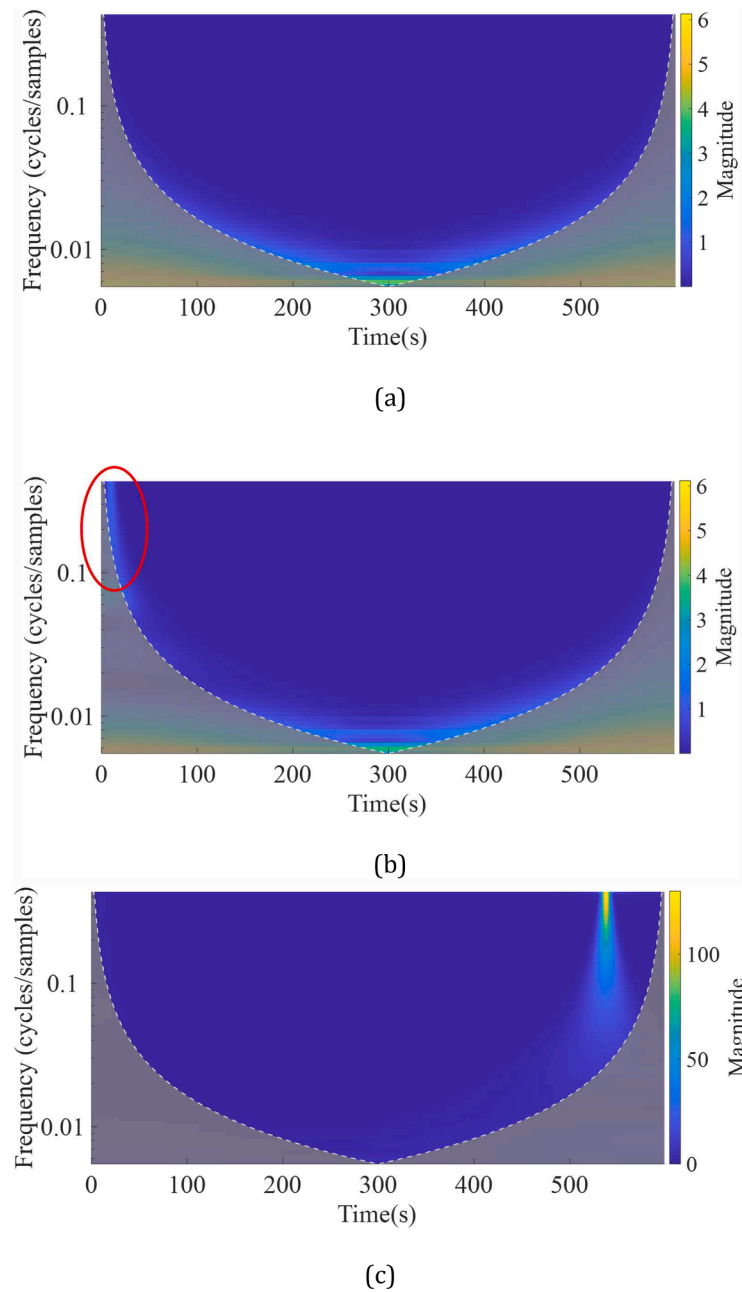


Fig. 12. Scalograms of the Sqnun (Status) signal under different operational scenarios (a) Normal-full load, (b) Disturbance-Bus failure, (c) Attack-denial of service.

windows. Visual inspection of the results listed in Table 5 reveals that the highest accuracy of the proposed approach is achieved when using Kaiser window. Therefore, Kaiser window is used in this work as it outperforms the other observation windows (i.e., Bartlett, Gaussian and Hamming).

4.2. The 4-IEDs Substation Test System

4.2.1. System Description

The second test system represents a 66/22kV substation with two transformers as illustrated in Fig. 8. The testbed was generated using Oracle VirtualBox and five more virtual machines (VMs); one VM simulated the primary plant using MATLAB, and the other four VMs used OpenPLC to represent different types of protection relays. In this work, cyber-attack cases consist of two attack types: FDIA attacks and Replay attacks. The FDIA violates the system's data to compromise its functionality and disrupt substation operation, while the replay attacks

intercept valid data transmission and fraudulently resend or delay it. These attacks could manipulate the circuit breakers' statuses causing serious implications for substation infrastructure and the system stability [44].

4.2.2. Detection Results

In this test system, the data in IED1 were used for training only while the data in IED2 were used for testing. The IED1 is located at transformer-1 and the IED2 is located at transformer-2 to provide overcurrent protection. The total number of GOOSE messages generated from IED1 is 16,605 packets including normal and attack cases, while the total number of GOOSE messages for IED2 is 8,645 packets. The implementation of the proposed approach leads to a detection accuracy of 99.72%, which demonstrates its efficacy in detecting the cyberattacks.

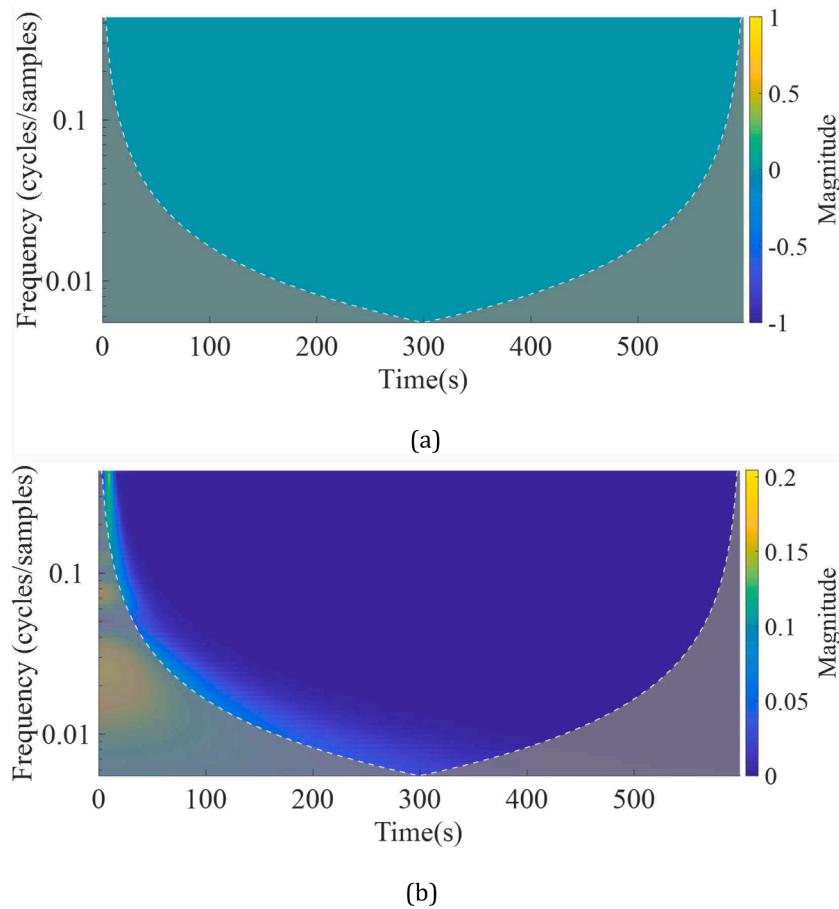


Fig. 13. Scalograms of the protection tripped signal under (a) Normal-full load and (b) Disturbance-breaker failure.

Table 6

Accuracy comparison with other detection approaches.

Study	Detection Approach	Accuracy (%)
[19]	CNN-LSTM and PSO	96.92
[20]	Deep capsule CNN	96.25
[46]	ML	94.00
[6]	MVCC	98.94
[7]	ARLS	95.00
[8]	GWO	98.63
[21]	BiLSTM	96.33
[13]	ML	97.00
[14]	ML	93.00
The Proposed approach	CWT-CNN	100

4.3. The Real-time Experimental Testing

The 4-IEDs system in (section 4.2) is modelled using OPAL-RT to evaluate the performance of the proposed approach in real-time.

4.3.1. Modelling the 4-IED Test System in OPAL-RT

Fig. 9 shows the experimental setup used in this work. The desktop computer (PC) represents the host and communicates with the OPAL-RT via Ethernet. The OPAL-RT's Real-time simulator (OP5033XG) hosts all the computations. The Field Programmable Gate Array (FPGA) is a configurable integrated circuit, that can send and receive signals and control field devices [45]. The General Electric Universal Relay (GE-UR) model is used to emulate the behaviour of the relay in real systems. The GE-Ethernet Switch model is Reason S20, and it complies with the IEC61850. The Global Navigation Satellite System (GNSS) precision time clock model is Reason RT430. This model is used to replicate the

behaviour of the GNSS precision time clock in a real-time environment. The 4-IEDs system is built in OPLA-RT using RT-LAB software and MATLAB. The IEC61850 communication protocol is used to establish the communications among the system IEDs, where GOOSE messages can be created and transferred between publishers and subscriber components in the system.

In the RT-LAB, the 4-IEDs system is split into two subsystems: The Graphical User Interface (GUI), which contains all system's inputs and outputs signals, and the other subsystem called the computation block where all signals can be compiled on a real-time environment as illustrated in Fig. 10.

4.3.2. Detection Results

In this testing system, the data in IED1 were used for training, while the data in IED2 were used for testing. The total number of GOOSE messages generated from IED1 is 14,453 packets, including normal and attack cases, while the total number of GOOSE messages for IED2 is 7,434 packets. The proposed detection approach was able to detect the cyber-attacks in this test system with a 99.45 % detection accuracy. This demonstrates its ability to detect the cyberattacks in real-time.

5. Discussion

In the 18-IEDs test system, the proposed approach used the combination of three features (i.e., Current, Stnum (Status), and Sqnum (Status)) for 4 IEDs (i.e., 10,12,22,21) as input data. This data will pass through the CWT to generate scalograms as illustrated in Figs. 11, 12, and 13. Fig. 11 shows the time-frequency scalograms of the current signal among different operational scenarios (i.e., normal-FL, disturbance-BF and attack-DM). These scalograms are obtained by applying

CWT to the IED's electric current data. Fig. 11 (a) clearly shows the changes in frequency scalogram colour, which reflects the normal changes in the IED current values in the time domain. Fig. 11 (b) represents a disturbance event under the breaker failure case. The light-yellow colour in the scalogram on the left-hand side refers to the point when the breaker failure occurred and hence leading to a rapid change in the data from the normal current values to zero. Fig. 11 (c) illustrates an attack event under a data manipulation scenario. The scalogram's colour looks similar to the normal case, except of the intensive yellow colour on the left-hand side of the scalograms. This colour refers to a malicious attack when the attackers spoof false information into the original current signal.

Fig. 12 depicts the time-frequency scalograms of the Sqnum (Status) signal among different operational scenarios (i.e., normal-FL, disturbance-BusF and attack-DoS). Fig. 12 (a) shows the scalogram under normal operation. The Sqnum is a counter that increases each time a GOOSE frame is sent. Since no events occurred during normal operation, no clear changes in the scalograms frequency colour appeared. Fig. 12 (b) represents a disturbance event under the busbar failure case. The light-yellow colour (highlighted in the red circle) in the scalogram on the left-hand side refers to the point when busbar failure occurred, which leads to a rapid change in the data from the normal current values to zero. Fig. 12 (c) illustrates the DoS attack. The scalogram's colour shows the intensive yellow colour on the right side of the scalograms. This colour refers to a malicious attack when the attackers deny the communication in the Sqnum counter.

Fig. 13 depicts the time-frequency scalograms of the protection tripped between two operational scenarios (i.e., normal-FL and disturbance-BF). Fig. 13 (a) shows the scalogram under normal operation. The Protection tripped value is a binary value equal to zero when there are no alarms, while it changes once alarms occur. During normal operation, the value of the protection tripped stayed zero all the time and therefore there are no changes in the scalogram magnitude, nor the frequency as illustrated in Fig. 13 (a). When the protection trip occurred in Fig. 13 (b) the value of this variable changed from zero to one. This change in the time domain value is represented in a change in the scalogram colour as shown in the light-yellow colour on the scalogram's left side.

5.1. Comparative Analysis with Other Detection Approaches

Table 6 lists the detection accuracies that are reported in the relevant studies in the literature. Visual inspection of the results reveals that the proposed detection approach was able to detect cyber-attacks and distinguish those attacks from normal and power disturbance cases at 100% accuracy. This is a significantly high detection accuracy compared with other reported detection accuracies in the literature. It is worth noting that the testing reported in the other studies in the literature was limited to only few specific types of attacks while the testing presented in this work involves six main attack types (Dos, MS, DM, FDIA, Replay, Composite attacks) that are considered the most popular attacks. Additionally, the proposed approach relied on only a subset of the features to perform the detection, which reduces the computational time. Furthermore, unlike the studies reported in the literature, the proposed detection approach in this paper was tested in real-time using OPAL-RT and it provides a detection accuracy of 99.45%. This demonstrates that the proposed approach is not only reliable and accurate in simulated systems, but also demonstrates its efficiency in a real-time testing system.

5.2. Computational Complexity

The proposed approach was executed on a computer with an Intel processor- Core (TM) i7-9750H CPU, 16.00 GB installed memory (RAM) and an operating system of 64-bit, Windows 10. The simulation of the test models is carried out on MATLAB/Simulink. During the model

training, the model will detect and extract the salient features from the input images (scalograms) to use them in the classification process. In the model testing, the trained network will be tested using new data instances not previously seen by the model, in order to evaluate the performance of the model based on test records that were correctly and incorrectly predicted by the model. The proposed approach requires a training time of 0.428 seconds and a testing time of 0.0983 second. This reflects how the proposed approach is effectively able to detect cyber-attacks and disturbance scenarios with less computational time and hence low complexity.

6. Conclusion

This research aims to detect cyberattacks that target smart digital substations and classify them from other power quality disturbances. In the current research, the proposed approach combines an enhanced continuous wavelet transform as a signal processing tool and a custom convolution neural network as a deep learning technique. First, the continuous wavelet transform is used to capture the abrupt changes in the data by generating the time-frequency scalogram. These images are then fed to a convolution neural network to extract and learn the features and patterns from these images and use them in the classification process. The physical and network features in IEC61850 are ranked to identify a subset of those features that strongly influence the detection classification accuracy of the cyberattacks. The proposed approach has been tested on two publicly available datasets. The first dataset consists of 18 IEDs and includes numerous attack types and various power disturbance events. The second dataset contains of 4 IEDs and includes two types of cyberattacks (FDIA and Replay attacks). In order to test the performance of the proposed detection approach, real-time testing using OPAL-RT was included in this work. The results have shown that the proposed approach was able to achieve 100% accuracy and 99.45% in real-time testing. The study has also identified the three key relevant features that strongly influence the cyberattack detection from disturbances and normal cases. Following the ranking of the physical and the network features, the study has shown that among all physical feature, the current is the most important feature for detection, while the Snum (Status) and the Sqnum (Status) are the most important features among the remaining network features.

Furthermore, the study looked into the effect of different observation windows on the detection accuracy of the proposed method. Specifically, the study investigated the effect of four observation windows namely Bartlett, Gaussian, Hamming and Kaiser. Since Kaiser window was able to provide the highest accuracy, the study concluded that Kaiser window is the most suitable observation window to use for the detection of cyberattacks.

Further research may look into implementing the proposed approach into other smart grid applications including the bulk systems as well as other energy sectors and non-energy sectors that are prone to cyber-attacks. Furthermore, the identification of the key relevant features to the cyberattacks opens the door for understanding how the cyberattacks may behave and hence identify the appropriate mitigating measure. Such an approach would enhance the cybersecurity of the smart digital substations.

CRedit authorship contribution statement

A. Abu Nassar: Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. **W.G. Morsi:** Conceptualization, Data curation, Funding acquisition, Project administration, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Data availability

The authors do not have permission to share data.

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